ISMIR 2026 · Abu Dhabi

How Much AI Is in This Track?

Quantifying the proportion of AI-generated stems in hybrid music mixtures

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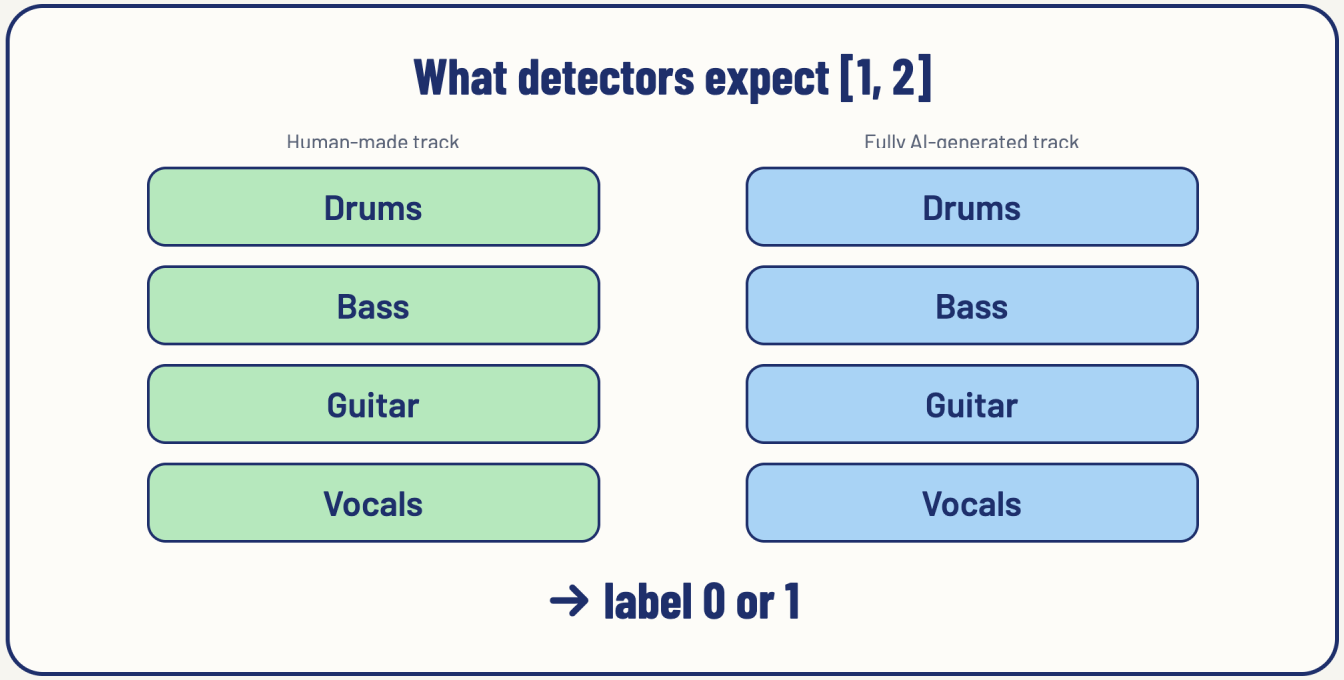
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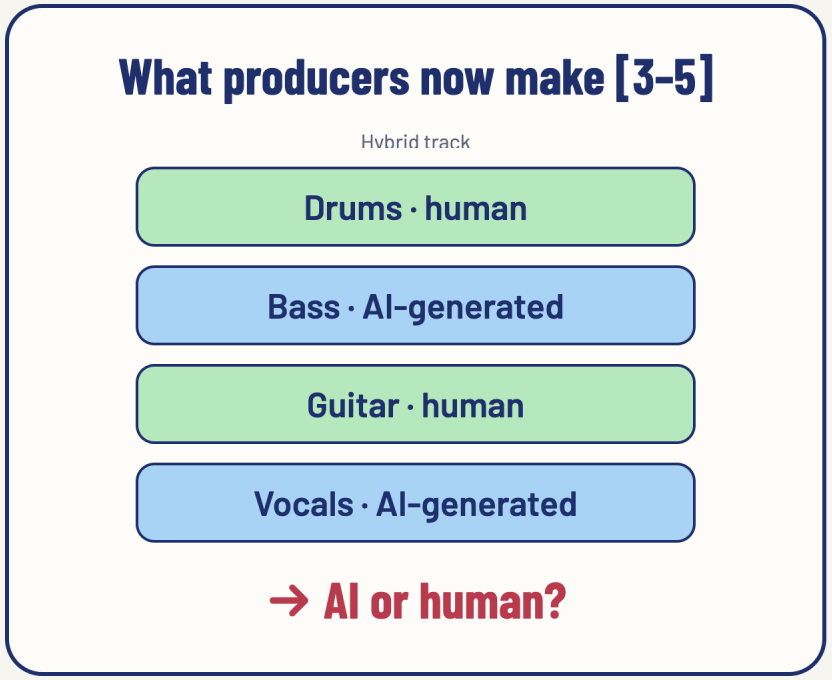
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Binary detectors assume a track is all AI or all human





[1] Afchar et al., "AI-generated music detection and its challenges," ICASSP 2025
[2] Rahman et al., "SONICS: Synthetic or not – identifying counterfeit songs," ICLR 2025
[3] Nistal et al., "Diff-A-Riff: Musical accompaniment co-creation via latent diffusion models," ISMIR 2024
[4] Pasini et al., "LiveBand: Live accompaniment generation in the audio domain," arXiv 2026
[5] Pons et al., "Music and artificial intelligence: Artistic trends," arXiv 2025

Binary is not enough

On pure tracks

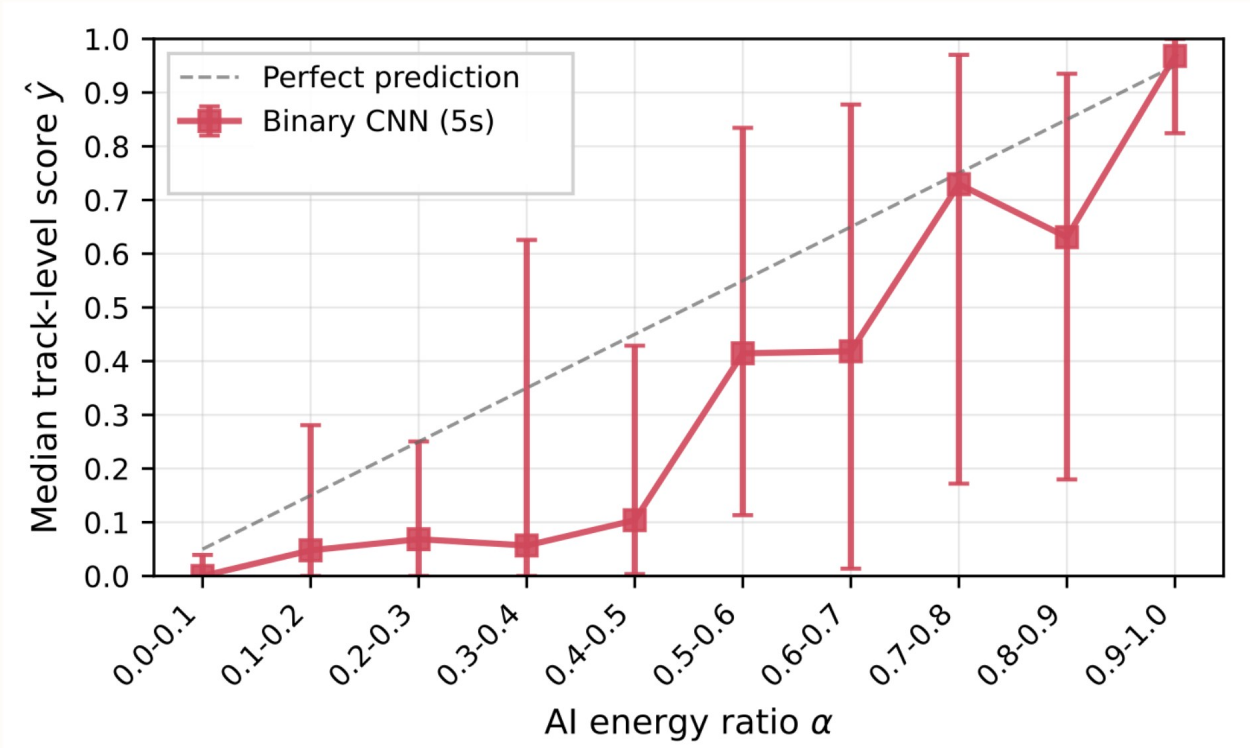
99.97%

accuracy of the reproduced detector [1]

On hybrid mixtures, score used as α

$R^2 = -0.85$

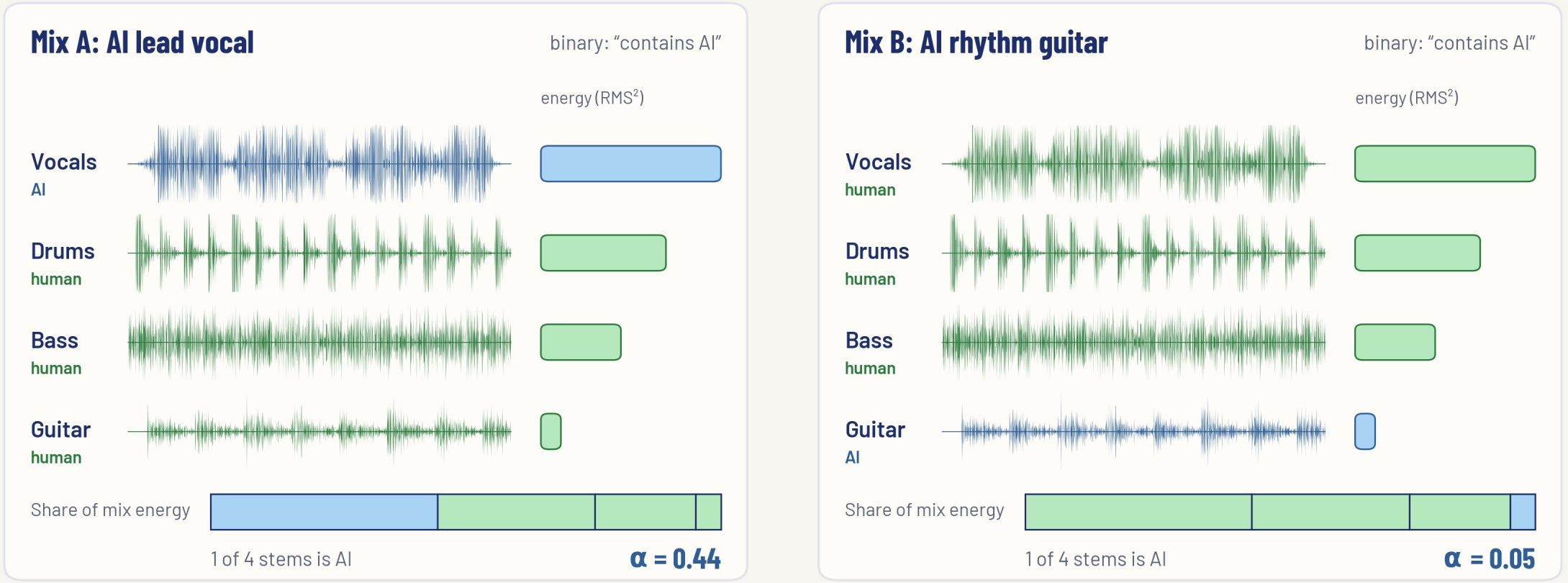
Median score stays near 0.05 until half the energy is AI



Held-out test split · 5-second windows · error bars: interquartile range

[1] Afchar et al., "AI-generated music detection and its challenges," ICASSP 2025

From "is it AI?" to "how much AI?"



Same stem count, same binary label, very different α: **each stem is weighted by its energy.**

Illustrative synthetic waveforms · α computed from each stem's RMS^2, as defined on the next slide

Problem formulation: regress the AI energy ratio α

$$x_{\text{mix}} = \sum_{i=1}^{n} s_i, \qquad s_i \in \mathcal{H} \cup \mathcal{A}$$

$$\alpha = \frac{\sum_{s_i \in \mathcal{A}} \text{RMS}(s_i)^2}{\sum_{i=1}^{n} \text{RMS}(s_i)^2}$$

α ∈ [0, 1]

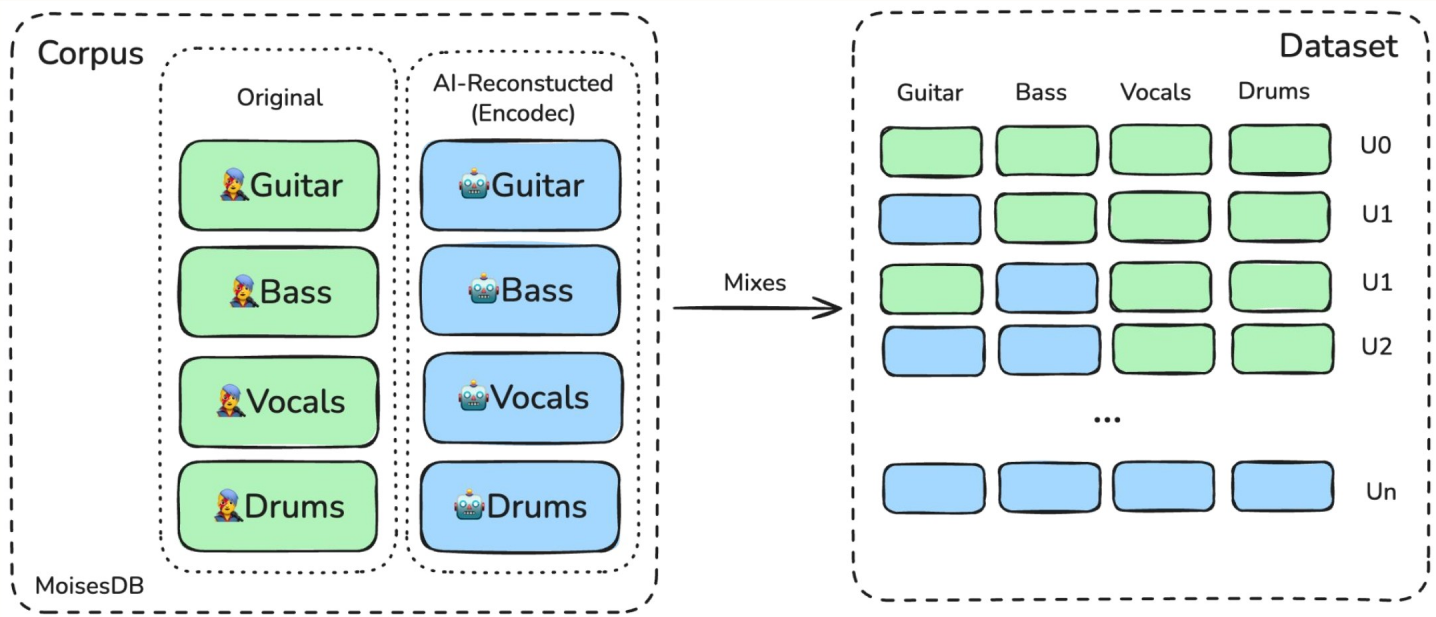
α = 0 for a fully human mix, α = 1 for a fully AI one

Model output

The model predicts $\hat{\alpha}$ directly from the mixed audio and reports "about X% AI"

$\mathcal{H}$: human-performed stems · $\mathcal{A}$: AI-generated stems · RMS: root-mean-square amplitude of each stem

ARIA: hybrid mixtures with known α



240
MoisesDB tracks [1]

2^n
real/AI combinations per track

21,212
mixtures with known α

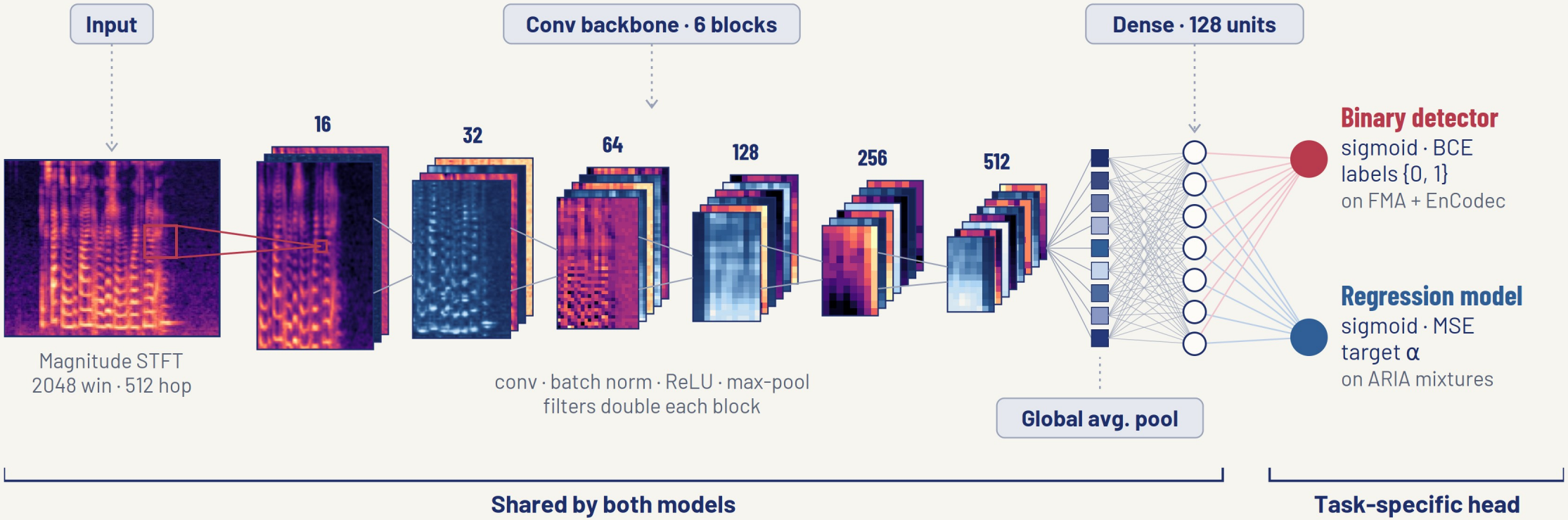
Encode–decode real stems with EnCodec [2] at 3 kbps: **same music, only the codec artifacts differ.**

ARIA = Authentic vs. Reconstructed Isolated Audio

[1] Pereira et al., "MoisesDB: A dataset for source separation beyond 4-stems," ISMIR 2023
[2] Défossez et al., "High fidelity neural audio compression," TMLR 2023

Same backbone, different head

Backbone of the reproduced binary detector [1]. Only the output layer and the loss change.

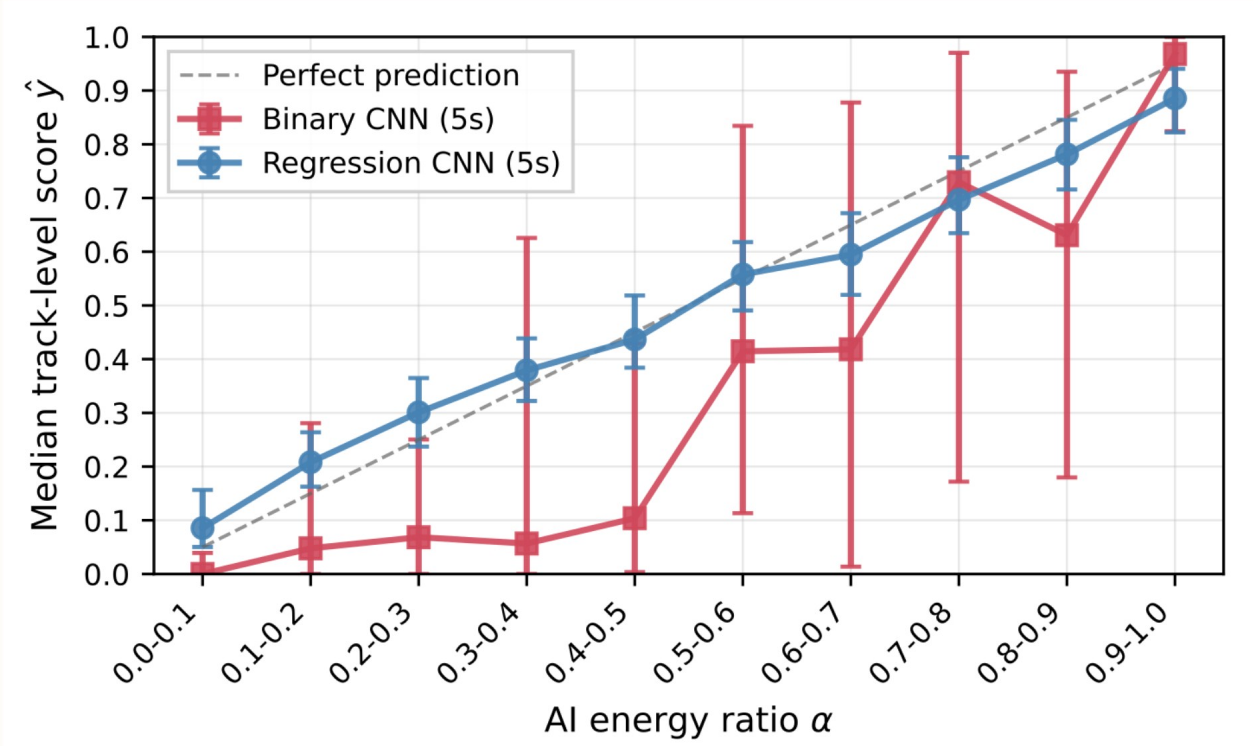


[1] Afchar et al., "AI-generated music detection and its challenges," ICASSP 2025
[2] Defferrard et al., "FMA: A dataset for music analysis," ISMIR 2017 · binary training data: FMA-Medium and its EnCodec reconstructions

Training on α recovers the proportion

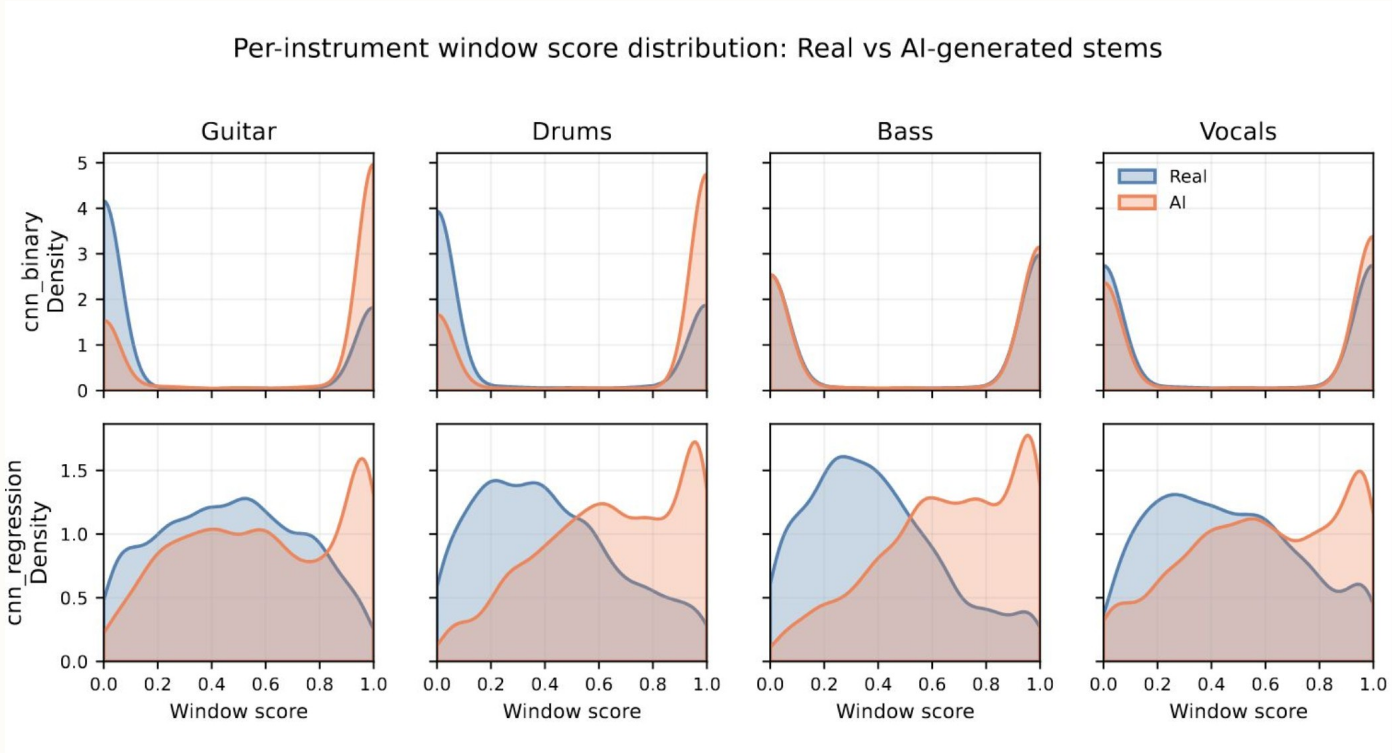
Model	MAE	R^2
Binary CNN score used as α	0.289	−0.85
Regression CNN	0.076	0.85

The proportional signal is in the mix. The training objective decides how well it is recovered.



Held-out test split: 24 tracks, 1,792 mixtures · 5-second windows, scores averaged per mixture

Detection sensitivity depends on the stem



Drums · Guitar

Binary scores separate cleanly: real near 0, AI near 1.

Vocals · Bass

Real and AI distributions largely overlap. An AI bassline can go unnoticed.

Regression

Partially separates vocals and bass.

Window-score densities, real vs. AI stem, test tracks with exactly vocals, drums, bass, guitar · 1-second windows · top: binary · bottom: regression

Takeaways

1

AI and human content are mixed

Producers combine AI-generated stems with human recordings.

2

Binary detection is not enough on its own

It works well on pure tracks but can't tell how much AI is in a mix.

3

Estimating the AI proportion is a first step

Training on α recovers it: MAE 0.076, R^2 0.85.

Not only “is it AI?” but “how much AI?”

In-distribution results: EnCodec 3 kbps reconstructions, held-out MoisesDB tracks

Limitations and next steps

Current scope	Next step
One codec at one bitrate: EnCodec, 3 kbps	Other codecs such as DAC [1], and commercial generators like Suno and Udio
AI stems are reconstructions of real recordings, not generated content	Paired real/AI stems from stem-conditioned generators [2, 3] or platform catalogs
No mastering or post-processing (EQ, compression, level balancing)	Production-realistic mixtures
Whole-mix, time-averaged estimate	Instrument-aware detection and temporal localization

[1] Kumar et al., "High-fidelity audio compression with improved RVQGAN," NeurIPS 2023
[2] Nistal et al., "Diff-A-Riff: Musical accompaniment co-creation via latent diffusion models," ISMIR 2024
[3] Pasini et al., "LiveBand: Live accompaniment generation in the audio domain," arXiv 2026

Thank you

Come see the poster

Paper: arXiv:2608.07285
Code and library: github.com/fergarciadlc/aria
fergarciadlc@gmail.com

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